

A Deep Learning Approach for Healthcare Insurance Fraud Detection

Precious Sihle Shungube

University of Johannesburg

Tebogo Bokaba

`tbokaba@uj.ac.za`

University of Johannesburg

Patrick Ndayizigamiye

University of Johannesburg

Siyabonga Mhlongo

University of Johannesburg

Eustice Dogo

Federal University of Technology Minna

Research Article

Keywords: Healthcare, Insurance fraud, Deep learning, Insurance Claims, Explainable AI

Posted Date: December 13th, 2024

DOI: <https://doi.org/10.21203/rs.3.rs-5453482/v1>

License: © ⓘ This work is licensed under a Creative Commons Attribution 4.0 International License.

[Read Full License](#)

Additional Declarations: No competing interests reported.

A Deep Learning Approach for Healthcare Insurance Fraud Detection

Precious Shungube¹, Tebogo Bokaba^{2*†}, Patrick Ndayizigamiye^{1†},
Siyabonga Mhlongo^{2†}, Eustace Dogo^{3†}

¹Centre for Applied Data Science, University of Johannesburg, Johannesburg, South Africa.

²Department of Applied Information Systems, University of Johannesburg, Johannesburg, South Africa.

³Department of Computer Engineering, Federal University of Technology, Minna, Nigeria.

*Corresponding author(s). E-mail(s): tbokaba@uj.ac.za;

Contributing authors: 218008217@student.uj.ac.za; pndayizigamiye@uj.ac.za;
siyabongam@uj.ac.za; eustace.dogo@futminna.edu.ng;

[†]These authors contributed equally to this work.

Abstract

Healthcare fraud is a global financial challenge affecting economic stability and trust in services, with traditional machine learning models struggling to accurately capture its complexity and adaptive nature. This study investigates the application of three deep learning (DL) models, which are artificial neural networks (ANN), convolutional neural networks (CNN) and long-short-term memory networks (LSTM) for healthcare fraud detection. This study used healthcare claim data, including patient demographics, claim amounts, diagnostic codes, and procedure types, to analyse healthcare service usage and identify fraudulent activity. To enhance the interpretability of these models, locally interpretable model-agnostic explanations (LIME) were used. The evaluation results demonstrated that the ANN was the best performer with an accuracy of 0.94, precision of 0.78, recall of 0.45, and F1-score of 0.57. While CNN excelled in accuracy, the LSTM was more effective in reducing false negatives. The LIME for ANN shows the prediction of a claim to be non-fraudulent with a high probability of 0.96, as opposed to a 0.03 probability of being fraudulent with ‘PotentialFraud’, as a driving feature, the evaluation metrics show that it is good at correctly identifying fraudulent cases. This study highlights the efficacy of integrating deep learning models with explainable AI (XAI), contributing to the growing research body in healthcare insurance fraud detection.

Keywords: Healthcare, Insurance fraud, Deep learning, Insurance Claims, Explainable AI

1 Introduction

Fraud poses a significant financial drain on healthcare systems globally, affecting not only insurance premiums but also healthcare resources [1]. The

intentional submission of fraudulent claims or falsifying facts to collect payments erodes financial stability and undermines trust in healthcare systems globally [2]. This pervasive issue leads to substantial increases in insurance premiums

and diverts funds away from much needed medical care, affecting families and economies on a considerable scale [3].

The urgency to address healthcare fraud has prompted the need for advanced technologies, notably DL, to detect fraud [4]. Such technologies are capable of sifting through complex datasets to identify or predict fraudulent patterns with much precision. Therefore, they have the potential to reduce fraudulent activities, thereby safeguarding the financial integrity of healthcare systems. Despite these advancements, deploying these technologies in fraud detection is not without challenges.

The effectiveness of ML techniques such as Naive Bayes (NB), random forest (RF), and gradient boosting in detecting fraud has been well documented. However, they sometimes struggle with the complexity and evolving nature of fraud schemes, particularly when dealing with high-dimensional and heterogeneous data sources [5]. On the other hand, deep learning techniques such as Recuring Neural Networks (RNNs), LSTM, and Graph neural networks (GNNs) are recognised for their ability to uncover intricate patterns and anomalies within extensive datasets [6]. The depth of learning and pattern recognition offered by these techniques enable a tailored response to the changing tactics of fraudsters. However, these techniques raise questions about their “black-box” nature, which may lead to lack the transparency required for both user trust and regulatory compliance [7].

Therefore, exploring deep learning in fraud detection requires balancing enhanced detection capabilities with the need for system transparency and simplicity. This paper investigates the use of DL in the context of fraud detection in insurance by analysing claims data from multiple healthcare providers. It aims to evaluate the performance of various DL techniques and identify the optimal model for detecting potential fraudulent healthcare insurance claims.

The paper also employs XAI to enhance the interpretability of the models. This paper offers practical insights on how deep learning could be used to curb fraudulent instances. The paper also offers theoretical contributions on application of XAI techniques to improve the interpretability of deep learning models, and the comparison of results from other studies.

The rest of this paper is structured as follows. Section 2 presents related works on healthcare fraud detection. The proposed methodology and its architecture are explained in Sect. 3, followed by results and the experimental studies in Sect. 4. Finally, limitations and the proposal of ideas for future research were mentioned, and conclude this paper in Sect. 5.

2 Related Literature

Healthcare fraud has emerged as a critical concern for the financial integrity of healthcare systems and insurance companies worldwide. According to Settipalli et al. [8], healthcare fraud not only leads to significant financial losses but also undermines trust in healthcare institutions and insurance providers. Given the scale of the insurance industry, which processes over \$1 trillion in premiums annually [9], the likelihood of fraudulent activities increases, necessitating advanced detection mechanisms. Fraud detection within this sector is made more challenging by the growing complexity and diversity of fraudulent schemes, which traditional rule-based systems are ill-equipped to handle [10]. Therefore, a pressing need exists for solid methodologies that can detect and mitigate fraud effectively in healthcare insurance claims.

Traditional fraud detection mechanisms have largely relied on rule-based systems, which, while effective to some degree, have proven insufficient in adapting to the evolving nature of fraud [11]. In earlier efforts, suspicious tendencies in healthcare fraud were primarily identified using predefined rules and expert knowledge [12]. However, the static nature of these systems made them vulnerable to the dynamic and sophisticated fraud schemes employed today. Rule-based systems often struggle with detecting subtle or emerging patterns of fraudulent activities, leading to missed opportunities for early detection. Consequently, the shift towards more flexible and adaptive systems like deep learning is essential.

A study by Yoo et al. [13] applied GNNs to detect Medicare fraud by analysing relationships among medical providers, beneficiaries, and physicians. The study indicated improved precision and recall in detecting Medicare fraud, suggesting that graph-based approaches are best suited for detecting fraud in healthcare claims [13]. However, while this study highlights the potential of GNNs, its

focus remains broad, and future research could benefit from narrowing the scope to specific entities, such as providers or beneficiaries, to optimise detection.

On the other hand, Zhang et al. [14] proposed a neural network-based approach for anomaly detection, focusing on disease-drug relationships. By quantifying these relationships and incorporating additional healthcare data, the model demonstrated improved performance in identifying medical fraud and abuse. The use of deep neural networks (DNNs) in this context has shown great promise in alleviating the workload of medical analysts while enhancing fraud detection capabilities. However, this method can be further refined by experimenting with various deep learning architectures and hyperparameter tuning to maximise detection accuracy and efficiency.

XAI offers a promising avenue for addressing the limitations of traditional fraud detection methods in healthcare. By making the decision-making process of artificial intelligence (AI) models more transparent, XAI can help us understand how these models arrive at their predictions [15]. This transparency is crucial in healthcare, where trust in AI-driven decisions is paramount. One of the key challenges in healthcare fraud detection is the complexity of the data involved. Traditional methods often struggle to capture the intricate relationships between different variables. XAI techniques, such as SHAP and LIME, can help to identify the most important features that contribute to a model’s predictions [16]. This information can be used to gain insights into the underlying patterns of fraudulent behaviour.

Moreover, XAI can aid in ensuring the fairness and accountability of AI-driven fraud detection systems [17]. Biases that may be present can be identified and mitigated by understanding the factors that influence a model’s decisions [8]. This is particularly important in healthcare, where disparities in access to care can exacerbate existing inequalities. In addition to improving transparency and fairness, XAI can also enhance the interpretability of AI models [18]. This can be beneficial for healthcare providers who need to understand the rationale behind a model’s recommendations. By providing explanations, XAI can foster trust and collaboration between humans and AI systems.

Various DL models, such as CNNs, LSTM networks, and GNNs have been applied to financial fraud detection. These models can effectively extract relevant features from structured and unstructured data, improving prediction accuracy. However, the lack of interpretability can make it difficult to understand the rationale behind these models’ decisions [7]. XAI techniques, offer a solution to this challenge by providing insights into the decision-making process of DL models [19]. Financial institutions can gain valuable insights into the patterns of fraudulent behaviour and develop more targeted detection strategies by understanding the factors that contribute most to a model’s predictions.

Key findings from [7] include the increasing use of CNNs for financial fraud detection, the emergence of GNNs as a promising approach for modelling network-based relationships, and the limited use of XAI techniques in this domain. Future research directions should focus on the application of XAI techniques to improve the interpretability of models.

3 Methods

This study employed deep learning techniques to detect healthcare fraud by analysing claims data from multiple healthcare providers. The primary objective was to classify whether a healthcare provider was involved in fraudulent activity, as indicated by the binary target variable “PotentialFraud”, which labelled providers as either fraudulent or non-fraudulent.

3.1 Experimental Setup

The setup was implemented using Python, a widely used programming language in data science which is praised for its efficiency and simplicity [20]. Python was used with essential libraries such as TensorFlow, Keras, and Scikit-learn for model development and evaluation. These libraries were designed to assist with creating and benchmarking new models [21]. The models were trained on a combination of inpatient, outpatient, and beneficiary datasets. The merged dataset was split into training and validation subsets using an 80/20 ratio. The 80/20 split is a common technique in machine learning and data science

for dividing a dataset into training and testing sets. This approach is often used because it provides a reasonable balance between sufficient training data and adequate testing data [22]. The experimental framework involved training multiple deep learning models, including ANN, CNN, and LSTM to predict the likelihood of fraudulent claims. The experimental setup focuses on detecting fraudulent behaviour in healthcare claims by applying deep learning models to the preprocessed dataset. The dataset was prepared by merging inpatient, outpatient, beneficiary, and provider data, which included important features such as claim amounts, deductible amounts, and demographic information. After preprocessing and merging the data, it was split into training and testing sets, where the training data was used to build the models and the testing data was reserved for model evaluation.

As shown in Figure 1, the proposed experimental framework for fraud prediction consists of seven main steps: data collection, preprocessing, data splitting, training, evaluation, parameter tuning, and using LIME for model interpretability.

3.2 Dataset

The data for this study was obtained from the Medicare Fraud Detection Dataset available on [23]. The dataset included multiple tables, Train and Test sets for claims data, as well as in-patient, outpatient, and beneficiary datasets. Beneficiary-data provides information about the beneficiaries, with 63,968 rows and 25 columns. Inpatientdata contains details of inpatient claims, with 9551 rows and 30 columns. Outpatientdata includes information on outpatient claims, with 125,841 rows and 27 columns. This comprehensive dataset provides a rich source of data for exploring the research questions. These datasets were merged on key identifiers such as ProviderID and BeneID to create a comprehensive view of each claim and its corresponding provider and beneficiary details. The merged dataset was then subjected to a series of preprocessing steps as shown in Figure 2.

Some of the features in the dataset are shown in Table 1.

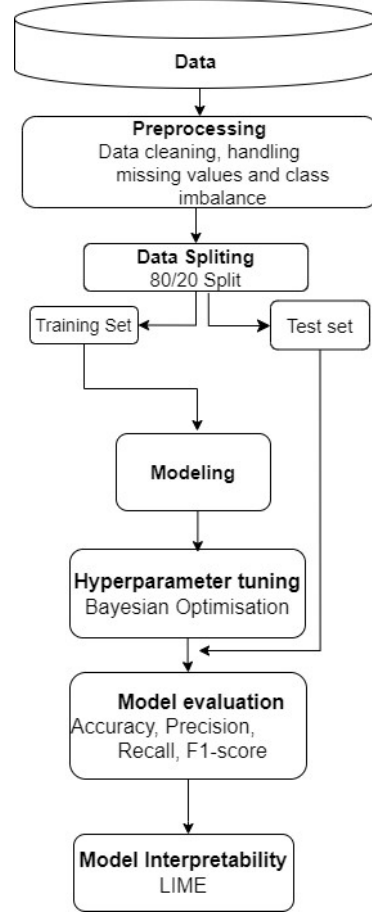


Fig. 1 Experimental Framework flow

3.3 Data preprocessing

Data preprocessing is an iterative process for the transformation of the raw data into understandable and useable forms [24]. It is the manipulation, filtration or augmentation of data before it is analysed, and is often an important step in the data mining process [25].

3.3.1 Handling Missing Values

In this study, addressing missing values was a critical step to ensure the integrity and completeness of the dataset. Initially, the dataset was examined and identified various features with missing values. To handle this efficiently, features were grouped based on their similarities. Grouping similar features helps in reducing the number of variables in the dataset by combining information from multiple related features. This can simplify the

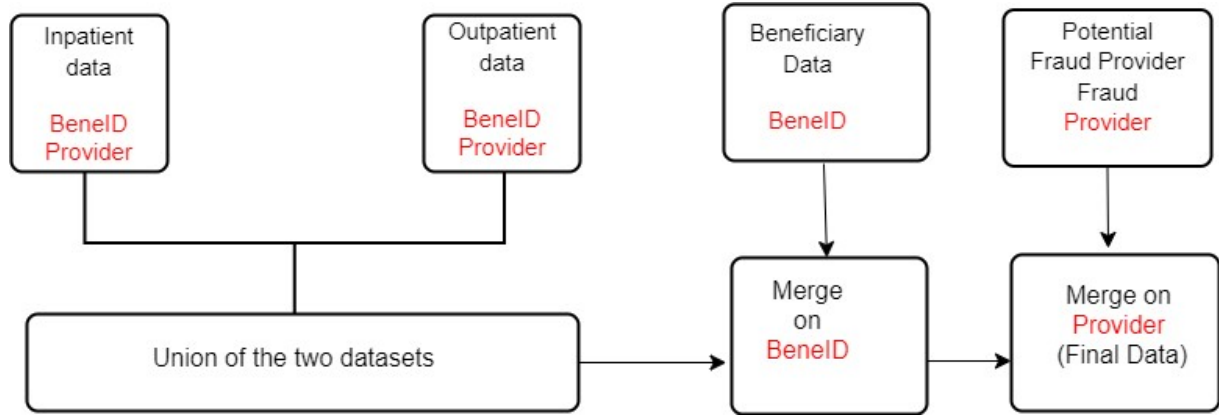


Fig. 2 Dataset Merging

Table 1 Dataset features

Feature	Description
Bene ID	Unique identifier of the beneficiary
DOB	Date of birth of the beneficiary
Gender	The gender of the beneficiary
Race	Human race of the beneficiary
State	State code in which a beneficiary belongs
Country	Country code in which a beneficiary belongs
Renal Disease Indicator	Indicates whether a beneficiary had a long history kidney disease
Chronic Condition Indicators *1,2,3,4,5,6,7,8,9,10,11	Represents a code which indicates whether a beneficiary had a chronic condition of a specific disease at the time of buying a plan
NoOfMonths.PartACov	Number of months of Part-A coverage.
NoOfMonths.PartBCov	Number of months of Part-B coverage.
IPAnnualReimbursementAmt	Maximum reimbursement amount allocated to a beneficiary for annual hospitalisation.
IPAnnualDeductableAmt	Maximum co-payment to be borne by the beneficiary annually if in case gets hospitalised.

Table 2 This table represents some of the features in the dataset and their description

analysis and reduce the risk of overfitting, especially when dealing with high-dimensional data [26]. This approach allowed for the combination of related features and derive new, informative features from these groupings. By doing so, the maximum amount of information was captured while consolidating and streamlining the feature set.

After creating new features from these groups, the redundant features with missing values were dropped as the new features were grouped based on these columns and derived maximum information from them. This ensured that the dataset

remained compact and focused, without losing valuable insights. By structuring the features in this manner, the interpretability and relevance of the dataset was maximised.

Upon completing this process, the dataset was re-examined for any remaining missing values. The result was a dataset with zero missing values, indicating that the handling of missing data had been successful and all necessary feature creations had been effectively implemented. This comprehensive approach improved the overall quality of the data

and provided a solid foundation for subsequent analysis.

3.3.2 Addressing Class Imbalances

In the dataset, a significant imbalance between fraudulent and non-fraudulent cases was encountered, with the latter vastly outnumbering the former as depicted in Figure 3. This imbalance can severely impact the performance of machine learning models, potentially leading to biased predictions that favour the majority class. To mitigate this issue, several strategies were employed, starting with stratification during the data splitting process. Stratification ensures representative sampling of classes in a dataset, particularly in imbalanced datasets [19]. By using the ‘stratify’ parameter in the ‘train_test_split’ function, both the training and validation sets maintained the same proportion of fraudulent and non-fraudulent cases as the original dataset. Despite this step, the distribution between classes remained skewed due to the substantial initial imbalance.

To further address this imbalance, the Synthetic Minority Over-sampling Technique (SMOTE) was implemented. SMOTE creates new data points for the minority class by interpolating between existing minority class points and their nearest neighbours [27]. As shown in Figure 4, It was particularly effective in oversampling the minority class (fraudulent transactions). This method generates synthetic examples of the minority class based on existing data points, thereby increasing the representation of fraudulent cases in the training set. This approach helped to balance the dataset more effectively than stratification, ensuring that the model had sufficient examples of fraudulent transactions to learn from without losing valuable information from the majority class.

3.4 Deep Learning techniques

Three deep learning models were applied to the preprocessed dataset to predict healthcare fraud: the ANN, CNN, and LSTM. Each model was trained and validated on a subset of the dataset, with their performance evaluated using accuracy, precision, recall, and F1-score metrics.

This study employed an ANN model, whose strength lies in its ability to learn from data [28].

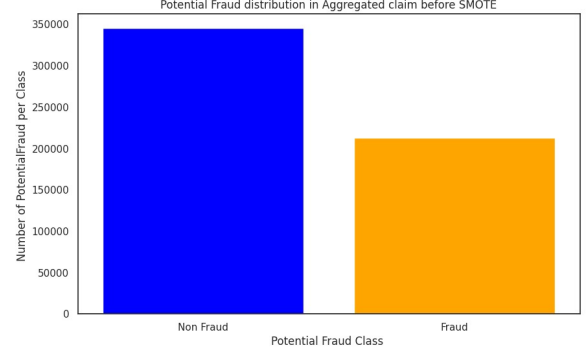


Fig. 3 Fraud classes before SMOTE

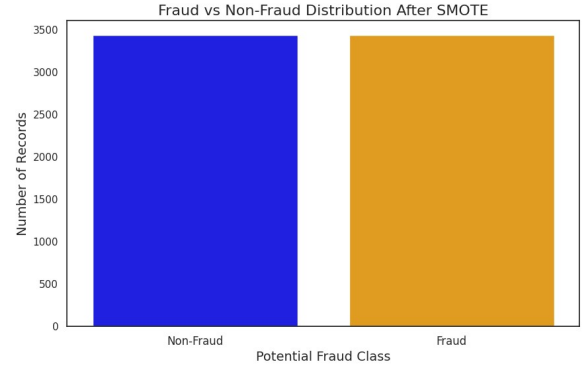


Fig. 4 Fraud classes after SMOTE

In order to reduce the discrepancy between its predictions and the actual outputs, the network modifies its internal parameters, known as weights and biases—after being exposed to a labelled dataset during the training phase [29]. The ANN model was structured with three hidden layers consisting of different neurons in each layer, both utilising ReLU activation functions. The output layer employed a sigmoid activation function, suited for the binary classification task of detecting fraud. The ANN was trained using the Adam optimiser with binary cross-entropy loss. The ANN model consists of multiple layers of neurons where the input layer, hidden layers, and output layer are fully connected [30]. The output from one layer is fed into the next layer, with activation functions applied to introduce non-linearity [31]. For each neuron, the formula is as follows:

$$z = \sum_{i=1}^n w_i x_i + b \hat{y} = \sigma(z) \quad (1)$$

where: - x_i is the input feature - w_i is the weight assigned to the feature - b is the bias term - σ is the activation function, such as ReLU for hidden layers and sigmoid for the output layer.

As shown in Equation (1), for each neuron, the weighted sum of the inputs is calculated, and an activation function, in this case, the sigmoid function, is applied to introduce non-linearity. This process is repeated through multiple layers, allowing the ANN to learn complex patterns and make accurate predictions

The CNN, a type of neural network that automatically learns important features from data using filters was used to capture patterns in the data by applying convolutional filters across the input features [32]. The CNN model included a one-dimensional convolutional layer followed by a max-pooling layer and dense layers. This model was particularly suited for identifying spatial relationships between features. The input data was reshaped to fit the requirements of the CNN architecture, and the model was trained using similar loss functions and optimisers as the ANN. The CNN uses convolutional layers to capture spatial relationships in data. The convolution operation is represented as:

$$z = (W * X) + b \quad (2)$$

Where: - $*$ denotes the convolution operation - W is the filter (or kernel) matrix - X is the input matrix - b is the bias term added to the result

As shown in Equation (2), the convolution operation involves applying a filter (W) to the input data (X) and adding a bias (b). This operation extracts features from the input data by sliding the filter across it. The result of the convolution is then passed through an activation function, such as ReLU, to introduce non-linearity. The max-pooling layer reduces the dimensionality of the feature maps while preserving the most important features. The dense layers then process the reduced feature maps to make a final prediction.

After convolution, the result is passed through an activation function such as ReLU:

$$\hat{y} = \text{ReLU}(z) \quad (3)$$

As shown in Equation (3), the ReLU function introduces non-linearity by outputting the maximum of 0 and the input value. Pooling layers

then reduce the dimensionality of the data using max-pooling or average-pooling operations.

The LSTM network was used to capture temporal dependencies within the data, such as sequences of claims over time. LSTM is a special kind of recurrent neural network designed to overcome a common issue called the vanishing gradient problem that occurs in traditional RNNs [33]. This problem makes it difficult for RNNs to remember information from the past when processing sequences of data. LSTM addresses this by using a unique memory cell structure that helps it retain important information over long periods [34]. LSTM models are well-suited for sequential data and were employed to leverage potential patterns in the temporal ordering of claims. The LSTM model used in this study had one LSTM layer followed by dense layers, with the Adam optimiser and binary cross-entropy loss applied for training. The LSTM model maintains a memory cell state and applies several gates (forget, input, and output) to update this state. The formulas for the LSTM cell are as follows:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (4)$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (5)$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \quad (6)$$

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t \quad (7)$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (8)$$

$$h_t = o_t \cdot \tanh(C_t) \quad (9)$$

where: - f_t , i_t , and o_t are the forget, input, and output gates

- C_t is the cell state

- h_t is the hidden state

- σ is the sigmoid activation function

- $\tanh$ is the hyperbolic tangent function

As shown in Equations (4) to (9), the LSTM cell uses the forget gate (f_t) to decide which information to discard from the previous cell state (C_{t-1}), the input gate (i_t) to determine which new information to store in the cell state, and the output gate (o_t) to control the information

flow from the cell state to the hidden state (h_t). The cell state (C_t) acts as a long-term memory, while the hidden state (h_t) captures short-term dependencies.

3.5 Evaluation Metrics

Each model was evaluated using several performance metrics, including accuracy, precision, recall, and F1-score. Accuracy was used to assess the overall correctness of predictions, while precision and recall focused on the model's ability to correctly identify fraudulent claims.

Accuracy is the most straightforward metric and is calculated as the ratio of correctly predicted instances (both positive and negative) to the total instances in the dataset. The formula for accuracy is shown in equation (10):

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (10)$$

where: - TP = True Positives (correctly predicted positive cases)
- TN = True Negatives (correctly predicted negative cases)
- FP = False Positives (incorrectly predicted positive cases)
- FN = False Negatives (incorrectly predicted negative cases)

Precision focuses on the accuracy of the positive predictions. Equation (11) shows that Precision is calculated as the proportion of true positive predictions to all predicted positive instances. The formula is:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (11)$$

Precision is important when the cost of false positives is high, ensuring that when the model predicts a positive class, it is likely to be correct.

Recall measures the model's ability to identify all relevant positive instances [35]. It is calculated as the proportion of true positives correctly identified out of all actual positive instances as shown in Equation (12). The formula for recall is:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (12)$$

Recall is critical in scenarios where missing positive instances (false negatives) are costly, such as in medical diagnoses or fraud detection.

As show in Equation (13), The F1-score provides a balance between precision and recall, particularly useful when there is an uneven class distribution. It is the harmonic mean of precision and recall, ensuring both metrics are equally weighted. The formula for F1-score is:

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (13)$$

The F1-score ranges from 0 to 1, with 1 being the best possible value. This metric is especially useful in imbalanced datasets, as it accounts for both false positives and false negatives [36].

3.6 Hyperparameter tuning

During the analysis, important variables that significantly influence the performance of a machine learning model include lag size, the number of hidden neurons, and epoch size. Tuning these parameters ensures the model's learning process is optimised for better accuracy, generalisation, and efficiency. The number of epochs was set to 10. The number of epochs refers to how many times the learning algorithm will work through the entire training dataset [37]. Too few epochs might lead to underfitting, where the model has not learned enough, while too many epochs may cause overfitting, where the model memorises the data rather than generalising well.

The Hyperparameter tuning technique that was used to tune parameters was the Bayesian optimisation. It helps reduce the time required to obtain an optimal parameter set [38]. In the initial iteration of Bayesian optimisation, hyperparameter values, sometimes referred to as data points, are selected at random. The choice of the point with the greatest degree of uncertainty in each cycle is then subject to trade-off with choosing a point from a region that currently produces the best outcomes is another name for this [39]. The results of the Bayesian technique are shown in Table 3 . These parameters were used the for the final models to achieve the target accuracy.

The ANN model contains three hidden layers: dense_93, dense_94, and dense_95. The respective number of neurons (or parameters) in each of these

Model	Accuracy	Best Parameters
ANN	0.94	Batch Size: 150 Num Neurons: 53
CNN	0.94	Batch Size: 120 Dense Units: 124 Kernel Size: 2 Num Filters: 48
LSTM	0.93	Batch Size: 33 Dense Units: 21 LSTM Units: 103

Table 3 Bayesian Hyperparameter Tuning Results

layers is 747, 3444, and 42. The number of hidden neurons in each layer impacts the model’s ability to capture complex patterns in the data [40]. More neurons allow for higher capacity and complexity, but can also increase the risk of overfitting if not properly regulated [41]. It is crucial to experiment with different values of hidden neurons to find the optimal balance between model complexity and performance. The CNN model contains four hidden layers: conv1d_15, max_pooling1d_15, flatten_15, dense_589, and dense_590. The respective number of neurons (or parameters) in each of these layers is 495, 0, 0, 4,378, and 23. The LSTM model contains three hidden layers: lstm_15, dense_621, and dense_622. The respective number of neurons (or parameters) in each of these layers is 151,240, 7,067, and 38. Tuning these parameters plays a pivotal role in enhancing model performance and reducing errors during prediction.

3.7 Model Interpretability

XAI is a crucial field that aims to make complex machine learning models more understandable to humans [42]. This is particularly important in domains like machine translation (MT), where understanding how a model arrives at its decisions can enhance trust and facilitate improvements.

LIME works by generating simplified local models around specific predictions. These local models are often linear or rule-based, making them easier to interpret than the original complex model. By examining the features that are most important in these local models, insights can be gained into how the original model made its decision for that particular input [43]. For example, in MT, LIME might reveal that certain words

or phrases played a more significant role in the model’s translation than others.

LIME works by perturbing the original input data to create synthetic samples that are close to the instance being explained. For each of these perturbed samples, LIME observes how changes in the input affect the model’s predictions. Using this information, LIME then builds a simpler interpretable model, such as a linear regression model, to approximate the behaviour of the complex model within the neighbourhood of the original instance. The perturbed samples are weighted based on their similarity to the original instance, with closer samples given more importance. This approach allows LIME to provide human-understandable explanations by highlighting which features were most influential in making the specific prediction. Through this process, users can understand why the model made certain decisions, identify possible model biases, and improve trust in the model’s predictions.

Although LIME is not based on strict mathematical formulas, the general approach can be seen as solving the following optimisation problem for the interpretable model:

$$\arg \min_{g \in G} \mathcal{L}(f, g, \pi_x) + \Omega(g) \quad (14)$$

- g is the interpretable model (e.g., linear or decision tree)
- G is the family of interpretable models
- f is the original complex model
- $\mathcal{L}$ is the loss function, representing how well the interpretable model g approximates f
- π_x is the proximity measure, defining how close the perturbed samples are to the original instance x
- $\Omega(g)$ is the complexity term, which penalises more complex models

This information can be used to identify areas where the model may be struggling or to improve the quality of future translations. By understanding the reasoning behind the model’s decisions, researchers can develop more accurate and effective MT systems. In addition, LIME can help to build trust between users and the AI system, as users can better understand how the model is making its predictions [44].

4 Results and Discussions

4.1 Exploratory Data Analysis

This study applied three deep learning algorithms to detect fraud in healthcare claims data, starting with an in-depth Exploratory Data Analysis (EDA) to identify key patterns and insights in the dataset. The dataset, consisting of merged inpatient, outpatient, and beneficiary data, provided a broad view of claims activity across numerous healthcare providers. Our initial EDA revealed several notable findings. As shown in Figure 5, the scatter plots presented highlight the relationship between the ‘IPAnnualDeductibleAmt’ and ‘IPAnnualReimbursementAmt’ across both fraudulent and non-fraudulent categories. The visualisations show two distinct subplots: one for the category ‘PotentialFraud’ = No (non-fraudulent cases) and another for ‘PotentialFraud’ = Yes (fraudulent cases). From the plot, it is apparent that there is no significant visible difference between the fraudulent and non-fraudulent groups in terms of the ‘IPAnnualDeductibleAmt’ and ‘IPAnnualReimbursementAmt’. In both categories, the values of the deductible and reimbursement amounts appear scattered within a similar range, without a clear pattern that distinguishes fraudulent claims from non-fraudulent ones based solely on these two variables.

This observation suggests that factors beyond just the deductible and reimbursement amounts may play a more significant role in identifying potential fraud, emphasising the need for further feature analysis or the inclusion of other variables in the predictive models. These results also stress the importance of employing more advanced statistical and machine learning techniques to uncover deeper insights into the fraudulent behaviours in this dataset.

Figure 6 highlights the distribution by state of beneficiaries, indicating that state 5 had the highest beneficiary concentration. This imbalance in geographic distributions may impact the generalisability of the models and suggest areas where targeted detection methods may be necessary.

Figure 7 implies that the dataset has a significant racial imbalance. Such imbalances can affect the performance of fraud detection models as the model may be biased towards patterns from

the overrepresented group. This could also potentially limit the generalisability of the model across diverse racial groups. From a research perspective, addressing this imbalance through techniques such as resampling or applying specific fairness constraints may be necessary to ensure that the model’s fraud detection capabilities do not favour any one racial group disproportionately.

In addition, the high percentage for race code “1” suggests that most fraudulent or legitimate healthcare claims are likely associated with this group. This could be a critical factor to consider when developing a deep learning model to detect healthcare fraud, as the model might learn patterns from this dominant group and perform poorly in underrepresented races, unless corrective measures are applied.

Figure 8 shows the distribution of attending physicians involved in healthcare fraud, with the orange bars indicating fraud, and the blue bars indicating non-fraudulent claims. A small subset of attending physicians is overwhelmingly responsible for a large number of fraudulent claims. This concentration suggests that fraud is not widespread across all providers, but rather isolated amongst a few. For example, physicians identified with the IDs “PHY139576” and “PHY150277” are disproportionately linked to fraudulent claims.

This insight highlights the importance of physician-level features in a deep learning model for fraud detection. The model can be trained to identify specific providers or identify patterns similar to those associated with known fraudulent actors. Moreover, this chart suggests that investigating and auditing a small subset of physicians could have a large impact on reducing fraud. Incorporating provider-level analysis could greatly improve the detection performance of fraud in the healthcare system.

The scatter plot in Figure 9 highlights an intriguing relationship between age and the amount of insurance claims reimbursed, with a clear distinction between fraudulent and non-fraudulent claims. In the top plot representing fraudulent cases, the observation shows that individuals within the age range of 30 to 70 years appear to engage in fraudulent behaviour more frequently. The scatter points are widely dispersed across the y-axis, indicating that fraud in this age group is associated with both low and high

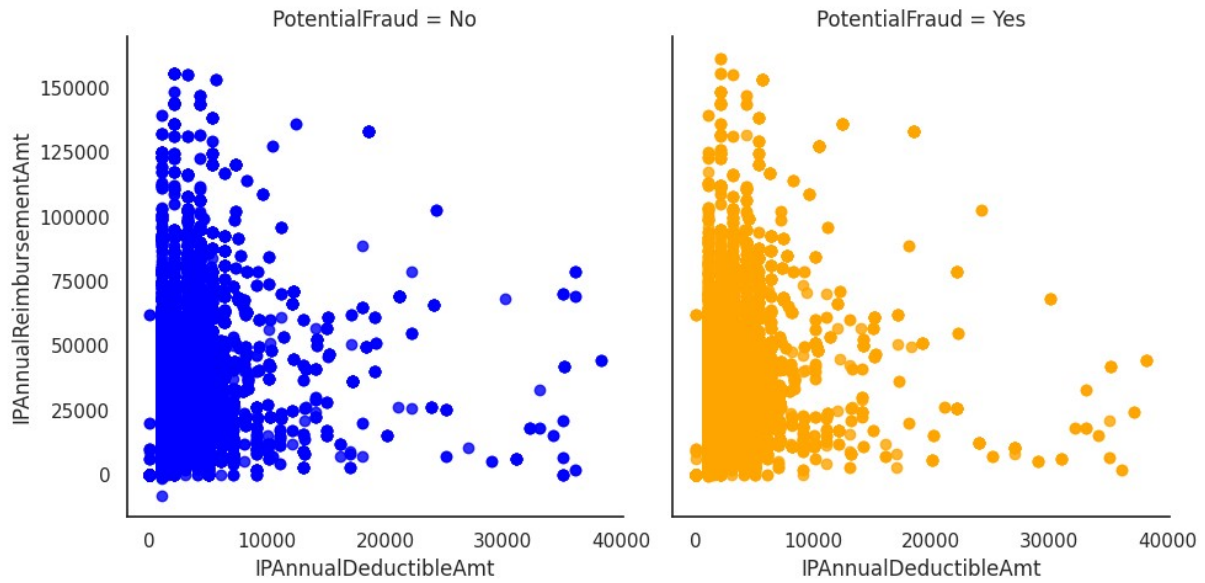


Fig. 5 Potential Fraud distribution in Aggregated claim transactional data

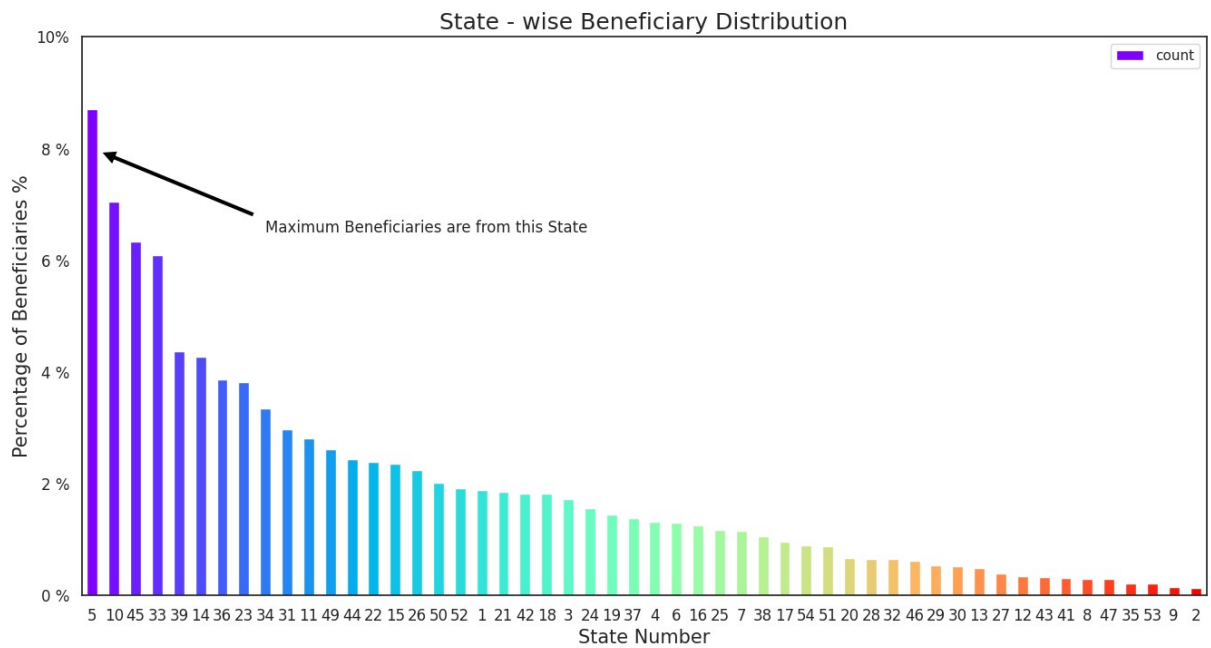


Fig. 6 Frequencies of Statewise beneficiaries

claim amounts. Notably, a considerable number of fraudulent claims surpass the 40,000 reimbursement mark, which suggests that larger claims tend

to be linked with fraudulent activity in these age ranges.

In comparison, the bottom plot illustrating non-fraudulent cases shows a more consistent and



Fig. 7 Race - wise Beneficiary Distribution

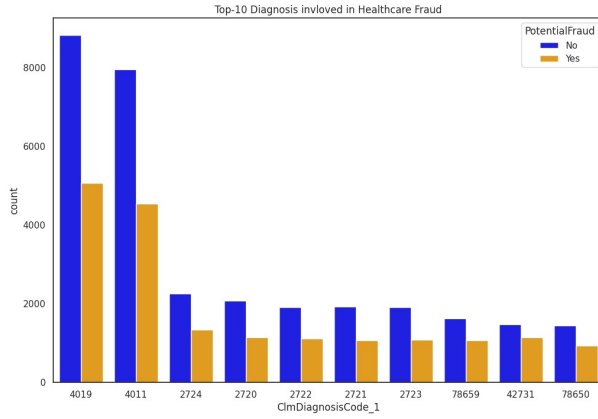


Fig. 8 Top-10 Diagnosis involved in Healthcare

concentrated pattern. Individuals of all ages, particularly those between 30 and 70 years, show fewer claims exceeding the 40,000 thresholds. The claims in this category are more tightly grouped below this mark, indicating that higher claim amounts are less common in non-fraudulent cases. This provides a valuable insight: in non-fraudulent behaviour, claim amounts tend to remain moderate, whereas fraudulent claims often involve higher financial reimbursements.

Moreover, age plays a critical role in distinguishing between fraudulent and non-fraudulent behaviour. Older individuals, particularly those above 70, show a lower incidence of fraud, as reflected by the sparse scatter points in the fraud plot for this age group. This suggests that fraud detection models could benefit from considering age as an important feature, as certain age

groups, like those between 30 and 70, are more prone to fraudulent behaviour, while older individuals seem less likely to be involved in such activities. The trends visible in the scatter plot indicate a correlation between age and fraudulent claims, particularly when large sums of money are involved.

4.2 Comparison of models performance

4.2.1 Performance metrics

After conducting the EDA, three deep learning models were applied, namely ANN, CNN, and LSTM. The ANN model achieved an accuracy of 94%, with a precision of 0.78, recall of 0.45, and an F1-score of 0.59. The CNN attained 93% accuracy, with a precision of 0.78, recall of 0.38, and an F1-score of 0.51. The LSTM, which captured temporal dependencies in the claims data, reached an accuracy of 93%, with a slightly higher precision (0.74) and a recall (0.47), resulting in an F1-score of 0.57.

Figure 10 shows that amongst these models, the ANN and LSTM emerged as the top performers in terms of overall accuracy, while the CNN demonstrated a higher precision, which makes it effective in reducing false negatives.

Precision, which measures the percentage of positive identifications that were actually correct, further highlights the strength of the ANN model, which outperforms all others. LSTM closely follows ANN in this regard, showing it is also effective in reducing false positives. CNN, while not as precise, still shows a relatively high precision, meaning that it accurately identifies a good portion of fraudulent claims.

The recall metric, which evaluates the ability of the models to correctly identify all fraudulent cases, also points to the effectiveness of the ANN and LSTM models. LSTM slightly edges out ANN in recall, reflecting its ability to capture a higher percentage of actual fraud cases, even if it occasionally includes non-fraudulent cases in its predictions. ANN shows moderate recall, indicating that it may miss some fraudulent claims.

The F1-score, a metric that balances precision and recall, reinforces the superior performance of the ANN and LSTM models. Both exhibit well-balanced scores, showing their effectiveness in

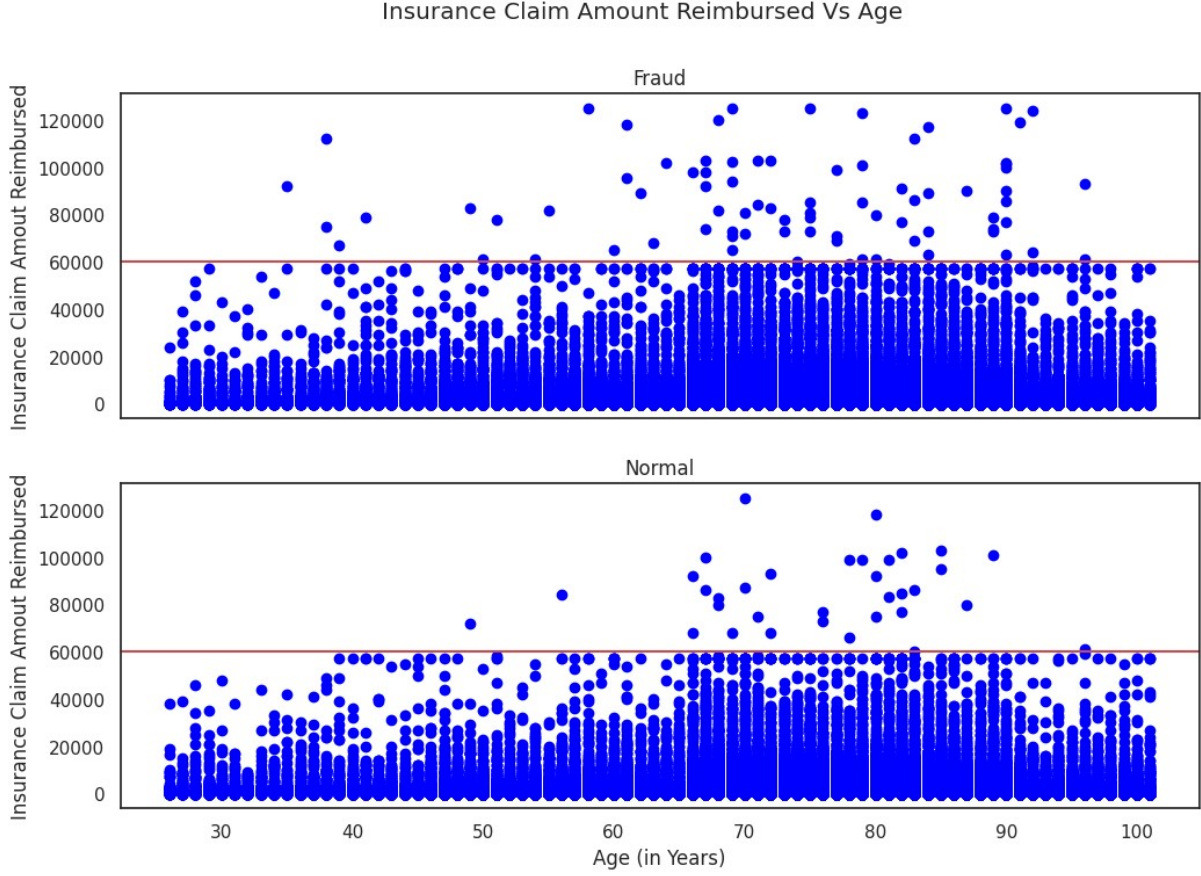


Fig. 9 Insurance claims Amount Reimbursed versus Age

accurately identifying fraud while also minimising false positives. The ANN follows with a decent F1-score, but its performance is not as balanced as the top two models.

Figures 11, 12, 13 show the confusion matrix of all three models. The CNN model correctly predicted 1455 instances of non-fraud and 58 instances of fraud, achieving an accuracy of 0.93. It performed well in predicting non-fraud instances but less well in predicting fraud instances. The ANN model correctly predicted 1451 instances of non-fraud and 69 instances of fraud, achieving an accuracy of 0.94. It performed well in predicting non-fraud instances but less well in predicting fraud instances. The LSTM model correctly predicted 1446 instances of non-fraud and 71 instances of fraud, achieving an accuracy of 0.93. It performed well in predicting non-fraud instances but less well in predicting fraud instances.

All three models achieved high accuracy, indicating overall good performance in terms of predicting non-fraud instances. However, in predicting fraud instances, the CNN model performed slightly better than the other models in terms of accuracy, recall and F1-score.

4.2.2 Comparison of the best performing model with existing literature

Table 4 summarises the performance of deep learning models from this current study and previous research, comparing their outcomes with the reference models used in each study. In this study, the best performing model, ANN, achieved a precision of 0.78 and a recall of 0.45, with an accuracy of 0.94. By effectively organising the features, notable performance improvements were observed.

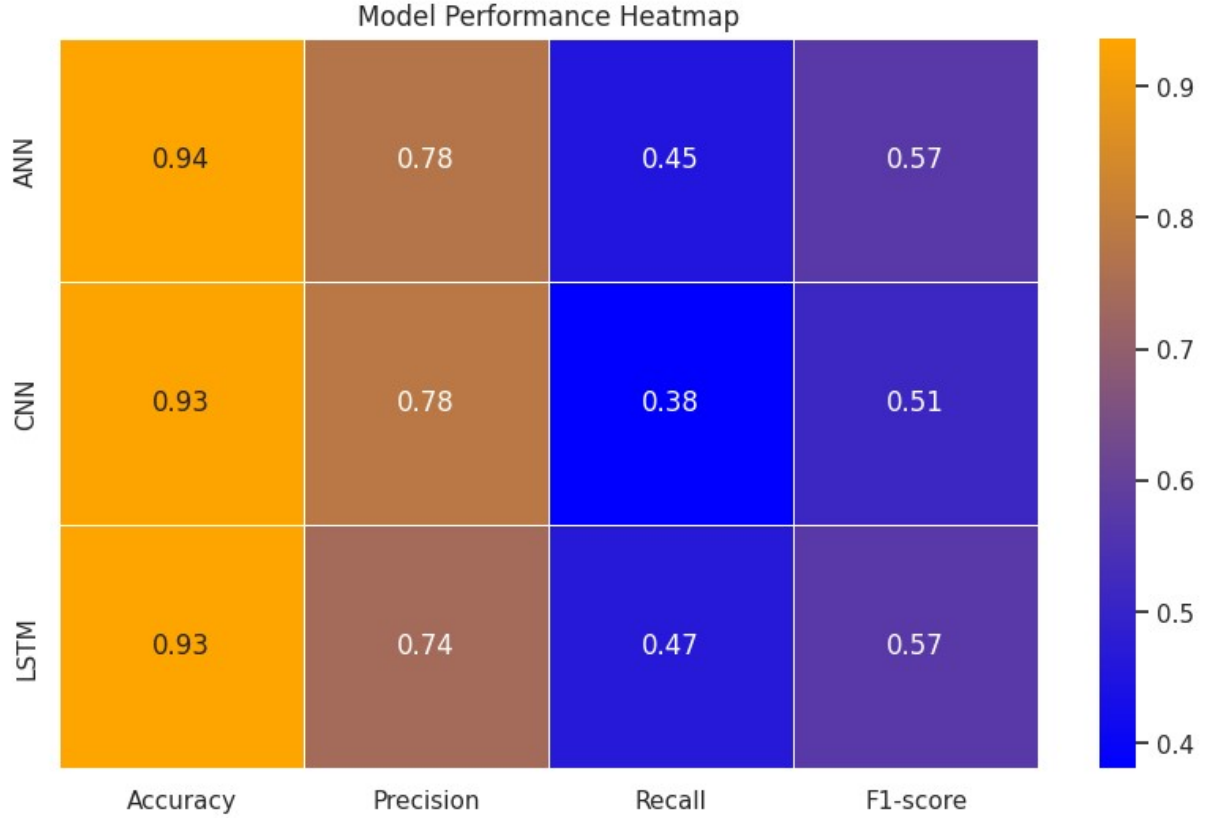


Fig. 10 Performance Comparison

Table 4 The summary of the performance difference.

Evaluation Metric	Gupta et al. (2021) [45]	Bounab et al. (2024) [46]	Yoo et al. (2023) [13]	Current Study
Accuracy	0.98	0.64	–	0.94
Precision	0.99	0.56	0.42-0.52	0.78
Recall	0.88	–	0.11-0.33	0.45
F1-score	0.93	0.69	0.19-0.39	0.57

When comparing the performance improvement of this study’s best performing model with other research on Medicare fraud detection, the current study generally exhibited superior gains in most metrics. Compared to a study using a GNN by Yoo et al. [46], this study’s ANN model demonstrated better performance improvements in precision, recall, and F1-score. The GNN model had its advantages in capturing relationships between medical providers and beneficiaries through graph analysis, but the current study’s

feature grouping combined with the CNN architecture produced higher performance in detecting fraudulent claims.

To address the issue of class imbalance, SMOTE was applied to balance the distribution of fraudulent and non-fraudulent cases. This improved this study’s model’s ability to differentiate between the two classes. In contrast, some studies, such as that of Gupta et al. [45], applied techniques such as subsampling and data embedding. Their neural network model reported higher

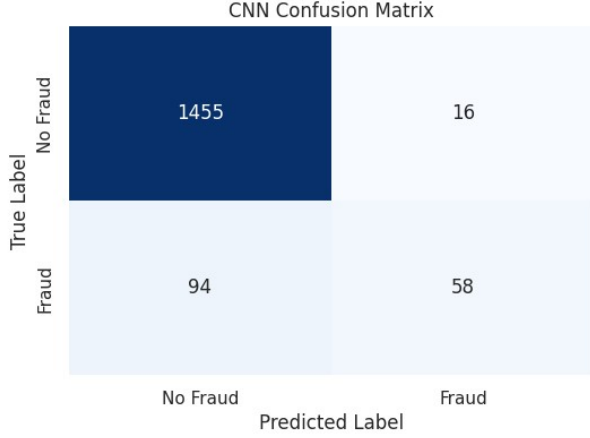


Fig. 11 CNN Confusion Matrix



Fig. 13 LSTM Confusion Matrix

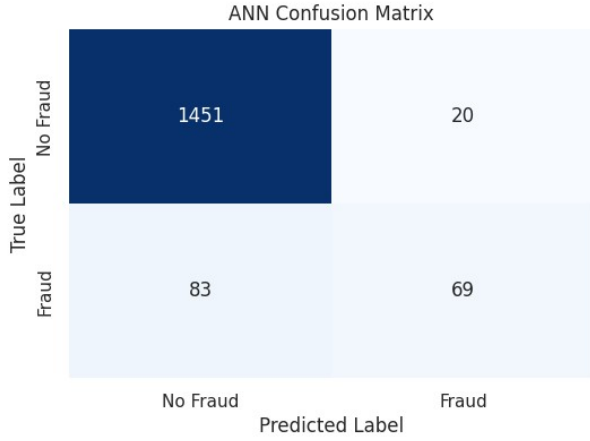


Fig. 12 ANN Confusion Matrix

F1-scores, likely due to the characteristics of the PM-JAY dataset that they used in India, which differs significantly from the Medicare dataset. Although their F1 scores were higher, the results of the current study on Medicare data remain strong and relevant, particularly in terms of accuracy and precision, as the model was carefully tailored to handle imbalances through SMOTE.

Bounab et al. (2024) [46] approached the class imbalance problem by employing a hybrid resampling technique, SMOTE-ENN, alongside Adaboost. Their study, focusing on the Medicare Part B dataset, highlighted the importance of handling noise in imbalanced datasets through the

combination of SMOTE and edited nearest neighbors (ENN). Adaboost, combined with this hybrid technique, allowed Bounab et al. [46] to achieve high scores, with decision trees (DT) outperforming other classifiers. Their emphasis on enhancing model performance through hybrid techniques resonates with the current study's use of SMOTE, further validating the importance of tailored solutions for imbalanced datasets.

In addition, unlike many previous studies that emphasised raw predictive performance, this study's approach integrated LIME, which provided interpretability alongside strong predictive metrics. This added layer of transparency is crucial in Medicare fraud detection, where understanding the rationale behind a model's decisions is as important as the accuracy of the predictions themselves. While Bounab et al. [46]. achieved excellent scores with Adaboost, their study did not address model interpretability to the same extent, underscoring a key strength of the current study's work in combining performance with explainability.

4.2.3 LIME Results

In the LIME plots in Figures 14, 15, and 16, the green bars represent features that contribute to the prediction of the positive class which is Fraud, while the red bars represent features that contribute to the negative class (non-fraud). The length of each bar indicates the strength of the contribution of the feature. Longer bars indicate features with a stronger influence on

the prediction, either towards fraud (green) or non-fraud (red). The green bars highlight features that increase the likelihood of the instance being classified as fraudulent, while the red bars show features that push the prediction towards non-fraudulent. For example, in this particular instance, features like “PotentialFraud” and “NumberOfMonths.PartACov” contribute positively towards a fraud prediction, whereas features such as “Provider” and “RenalDiseaseIndicator” contribute towards a non-fraud classification.

The LIME output for the ANN model provides valuable insights into how the model arrived at its decision for predicting whether a claim was fraudulent or non-fraudulent. The results as shown in Table 5 indicate that the ANN model was quite confident in predicting this instance as non-fraudulent, with a high probability of 0.96, compared to a probability of 0.04 for fraud. This confidence in the non-fraud prediction suggests that the features selected by the model strongly aligned with patterns typically associated with legitimate claims. Although the ANN has a slightly low probability of fraud detection, its precision of 0.78 and accuracy of 0.94 is leading among the other models, which means it is good at correctly identifying all fraudulent cases.

Figure 14 shows that ‘Potential Fraud’ was one of the most influential features in the prediction model. The ANN model relied heavily on this feature to determine that the claim was likely fraudulent. The ‘PotentialFraud’ feature showed a significant positive influence towards fraud, indicating that this claim exhibited characteristics associated with fraudulent behaviour.

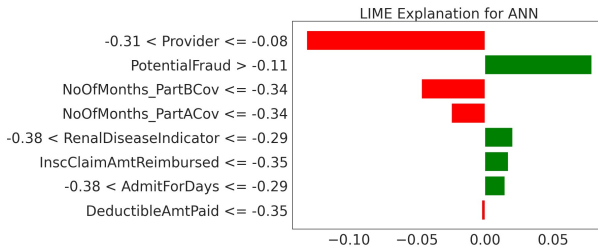


Fig. 14 LIME for ANN

Other important features such as ‘NoOfMonths.PartACov’ and ‘NoOfMonths.PartBCov’,

which represent the number of months the beneficiary was covered under Part A (NoOfMonths.PartACov) and Part B (NoOfMonths.PartBCov), respectively, also played a substantial role in driving the prediction toward non-fraud. This suggests that beneficiaries with longer coverage histories are generally associated with legitimate claims in the dataset. The ‘AdmitForDays’ feature, indicating that the number of days the patient was admitted, contributed marginally towards fraud detection, suggesting that extended hospital stays might raise suspicion, although this effect was outweighed by other factors. The insurance claim amount reimbursed (‘InscClaimAmtReimbursed’) and the number of days admitted (‘AdmitForDays’) also had notable influences, leaning towards the fraudulent outcome, indicating that the amounts involved were consistent with abnormal claim patterns and the number of days that the patient was admitted contributed marginally to fraud detection. This comprehensive analysis using LIME reveals how various features collectively influence the ANN model’s decision-making process, enhancing the interpretability of complex machine learning models in fraud detection.

The LIME output for the CNN model offers insights into how this model interpreted the features to classify the claim as non-fraudulent. In this instance, Table 5 shows that CNN predicted the claim to be non-fraudulent with a high probability of 0.97, as opposed to a 0.03 probability of being fraudulent. This level of confidence in predicting non-fraud reflects the model’s reliance on particular characteristics that align more with nonfraudulent behaviour. The CNN’s probability of predicting fraudulent instances is lower than the ANN and LSTM.

Figure 15 shows that the key feature driving the fraud prediction is ‘PotentialFraud’. This indicates that the model used the provider’s identity and associated historical data as a key determinant in assessing the legitimacy of the claim. Features like ‘NoOfMonths.PartACov’, representing the beneficiary’s Part A coverage duration, also played a significant role in guiding the model towards a non-fraudulent decision. This suggests that beneficiaries with extended coverage histories tend to have a lower likelihood of submitting fraudulent claims, according to the CNN model’s learned patterns. The ‘AdmitForDays’ feature had

a minor influence towards fraud but were ultimately outweighed by the dominant non-fraud signals. The ‘InscClaimAmtReimbursed’ value supported the non-fraud classification, indicating that the financial aspects of this claim were within the expected range for legitimate claims. Overall, the LIME output for the CNN model demonstrates how a combination of these features influenced the model’s interpretation, highlighting the importance of certain attributes like ‘Potential-Fraud’ in distinguishing between fraudulent and non-fraudulent claims.

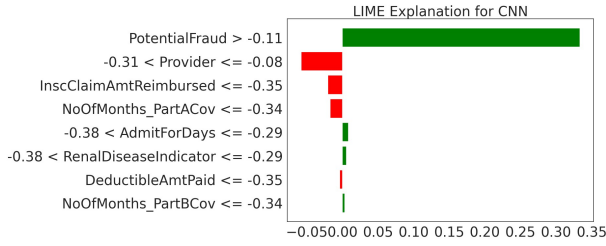


Fig. 15 LIME for CNN

In Table 5, the LSTM predicted that the claim was non-fraudulent with a probability of 0.95, compared to a very low probability of 0.05 for fraud. This high confidence in predicting non-fraud indicates the model’s reliance on a set of features that strongly align with legitimate claims. The LSTM achieved the highest probability of predicting fraudulent instances.

Table 5 Prediction Probabilities from LIME

Model	Non-Fraud	Fraud
ANN	0.96	0.04
CNN	0.97	0.03
LSTM	0.95	0.05

Figure 16 shows that the ‘Provider’ feature emerged as one of the most influential factors driving the model’s non-fraudulent prediction. This suggests that, similar to other models, the LSTM heavily relied on the historical behaviour and patterns of the specific provider to determine the legitimacy of the claim. The model identified this provider as having a history consistent with non-fraudulent claims, which contributed

significantly to the high probability assigned to the non-fraud class. The ‘PotentialFraud’ feature, which is a key indicator of fraudulent activity, also contributed to the non-fraud prediction but to a lesser extent than the ‘Provider’ feature. Financial attributes, such as ‘InscClaimAmtReimbursed’ (the reimbursed insurance claim amount) and ‘DeductibleAmtPaid’, further reinforced the model’s decision towards non-fraud. These features likely fell within ranges typical for legitimate claims, influencing the model’s confidence in this classification.

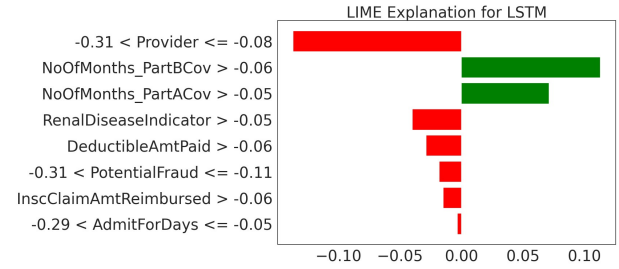


Fig. 16 LIME for LSTM

Interestingly, non health-related indicators like ‘NoOfMonths.PartA.Cov’ and ‘AdmitForDays’ (number of days admitted) also played a role in the LSTM’s decision-making. Although they showed some influence towards fraud, the overall impact of these features was outweighed by the dominant non-fraud indicators. The ‘NoOfMonths.PartA.Cov’ and ‘NoOfMonths.PartB.Cov’, which represent the duration of Part A and Part B coverage, respectively, supported the non-fraud prediction. This suggests that beneficiaries with prolonged insurance coverage were more likely to submit legitimate claims according to the model’s learned patterns.

The LIME interpretation for the LSTM model indicates that it effectively leveraged a combination of provider history, financial claim attributes, and health-related factors to differentiate between fraudulent and non-fraudulent claims. This reinforces the model’s capability to process temporal dependencies and patterns, as expected from an LSTM network, in understanding complex fraud detection scenarios within the healthcare claims data. The output highlights the transparency and

interpretability that LIME brings to understanding how deep learning models like LSTM make predictions, which is crucial for building trust in automated fraud detection systems.

The comparison of the performance metrics and LIME interpretations across the ANN, CNN, and LSTM reveals distinct advantages and shortcomings for each model in the context of healthcare fraud detection. From this analysis, CNN emerges as the top-performing model based both on its quantitative performance metrics and its interpretability through LIME.

In terms of performance metrics, the ANN shows slightly better accuracy, precision, recall, and F1-score compared to the other models. Its accuracy of 0.94 reflects a strong overall performance in distinguishing between fraudulent and non-fraudulent claims. More importantly, its precision of 0.78 and recall of 0.45 suggest that it maintains a good balance between avoiding false positives (wrongly classifying legitimate claims as fraud) and successfully identifying actual fraud cases. This balance is critical in fraud detection, where incorrectly flagging legitimate claims can lead to unnecessary investigations and increased costs, while failing to detect fraudulent claims can result in financial losses for healthcare providers. The higher F1-score of the ANN (0.57) compared to the CNN (0.51) indicates that it provides a more reliable overall performance, especially in contexts where both precision and recall are crucial.

Looking at interpretability, LIME analysis provides valuable information on how each model arrives at its predictions. For CNN, LIME highlights key features like 'PotentialFraud', and ('AdmitForDays') as primary drivers in determining fraud. The CNN effectively leverages these features to build its decision boundaries, showing a high probability for predicting legitimate claims as non-fraudulent, while still capturing the subtleties of fraudulent patterns when necessary. The fact that the ANN can handle complex feature interactions more effectively than the other models, as evidenced by the LIME analysis, further enhances its value. This is particularly important in fraud detection, where patterns of fraudulent activity can be intricate and multi-faceted, involving combinations of billing amounts, provider histories, and patient characteristics.

In contrast, while the CNN and LSTM models demonstrate solid performance, they exhibit certain limitations that prevent them from outperforming the ANN. The CNN, while achieving a high accuracy of 93%, struggles slightly with recall (0.38), indicating a higher tendency to miss actual fraud cases. The LIME analysis for CNN suggests that its decision-making is heavily influenced by the 'PotentialFraud' feature, but the model's slightly lower recall suggests it may not capture the full complexity of fraudulent claims as effectively as the ANN. The LSTM, while adept at handling temporal dependencies in the data, exhibits even lower recall (0.47), meaning it is struggling at capturing more instances of fraud. The LSTM's reliance on features like the number of days spent in hospital ('AdmitForDays') and billing-related features like 'NoOfMonths_PartBCov' shows that it captures important fraud-related patterns but may be less effective in identifying more nuanced or subtle cases of fraud, which can manifest in complex feature interactions.

The combination of deep learning models, EDA, and visualisations provided a comprehensive approach to detecting fraud in healthcare claims data. The models achieved high accuracy, and the visualisations enhanced the interpretability of the results, helping to uncover specific procedures, diagnosis codes, and demographic characteristics that were strongly associated with fraud. These findings offer valuable insights that can guide the development of more targeted and effective fraud detection strategies in the healthcare industry.

5 Conclusion

This study successfully applied deep learning techniques in the context of detecting fraudulent and non-fraudulent healthcare insurance claims. By analysing comprehensive data from healthcare providers, key features that were critical in detecting fraudulent claims were identified, such as age, reimbursement amounts, number of claims per provider, and specific procedures. Various deep learning models, including ANN, CNN, and LSTM, were trained and evaluated on these features.

The results prior to the introduction of LIME demonstrated that the CNN model was particularly effective at detecting non-fraudulent claims, while LSTM excelled in identifying fraudulent

ones. The ANN was quite good at detecting fraud with a precision of 0.78 and accuracy of 0.94 which is leading among the other models. The integration of explainable AI methods, such as LIME, allowed us to further interpret the models' predictions, highlighting what specific features the models relied on to classify claims as fraudulent or non-fraudulent. This post-LIME analysis confirmed the strengths and limitations of each model, providing valuable insights into how these models can be applied in practical fraud detection systems.

Importantly, before model training was performed, key data challenges such as missing values and class imbalances were addressed using the SMOTE technique. This step was important to ensure the strength and reliability of the results. However, this study is not without limitations. The data only contained race codes rather than actual race information, which limited the model's ability to detect potential fraud patterns that might be influenced by racial demographics. In addition, the use of procedure codes instead of detailed procedure descriptions restricted the granularity of analysis. The actual procedures could provide more specific insight into potentially fraudulent behaviours, as certain procedures are more likely to be associated with fraud than others. Addressing these limitations with more detailed and comprehensive data could further improve the accuracy and applicability of fraud detection models.

Building on this study, future research could explore several key areas to improve healthcare fraud detection further. First, incorporating more granular data, including detailed procedure descriptions and socioeconomic information, could enable the models to identify nuanced fraud patterns. In addition, integrating data from external sources, such as government and public health records, may provide valuable context that could improve the accuracy of the model. Future studies could also experiment with hybrid model architectures that combine the strengths of CNN, LSTM, and other deep learning techniques to improve recall rates, especially to identify complex fraud cases. Furthermore, real-time fraud detection could be explored by adapting these models to detect suspicious claims as they are processed, allowing healthcare providers to intervene promptly.

This study contributes to the growing body of research on the application of deep learning in fraud detection, particularly within the healthcare sector. Practically, it offers a framework for healthcare providers and insurance companies to enhance their fraud detection efforts, potentially saving substantial financial losses through more accurate detection of fraudulent claims. The findings of this study highlight the importance of continuous research and development in the field of fraud detection. By utilising deep learning models, integrating explainable AI techniques, and incorporating additional data sources such as government records and socioeconomic factors, more robust fraud detection systems that are well-suited for real-world applications can be created.

References

- [1] Hyman, D.A.: Health care fraud and abuse: market change, social norms, and the trust "reposed in the workmen". *The Journal of Legal Studies* **30**(S2), 531–567 (2001) <https://doi.org/10.1086/324674>
- [2] Morris, L., Thomspson, G.W.: Reflections on the government's stick and carrot approach to fighting health care fraud. *Ala. L. Rev.* **51**, 319 (1999)
- [3] Li, J., Huang, K.-Y., Jin, J., Shi, J.: A survey on statistical methods for health care fraud detection. *Health care management science* **11**, 275–287 (2008)
- [4] Bouchetara, M., Zerouti, M., Zouambi, A.R.: Leveraging artificial intelligence (ai) in public sector financial risk management: Innovations, challenges, and future directions. *EDPACS* **69**(9), 124–144 (2024) <https://doi.org/10.1080/07366981.2024.2377351>
- [5] Aslam, N., Khan, I.U., Mirza, S., AlOwayed, A., Anis, F.M., Aljuaid, R.M., Baageel, R.: Interpretable machine learning models for malicious domains detection using explainable artificial intelligence (xai). *Sustainability* **14**(12), 7375 (2022) <https://doi.org/10.3390/su14127375>
- [6] Choi, K., Yi, J., Park, C., Yoon, S.: Deep

- learning for anomaly detection in time-series data: Review, analysis, and guidelines. *IEEE access* **9**, 120043–120065 (2021) <https://doi.org/10.1109/ACCESS.2021.3107975>
- [7] Kute, D.V., Pradhan, B., Shukla, N., Alamri, A.: Deep learning and explainable artificial intelligence techniques applied for detecting money laundering—a critical review. *IEEE access* **9**, 82300–82317 (2021) <https://doi.org/10.1109/ACCESS.2021.3086230>
- [8] Settipalli, L., Gangadharan, G.: Wmtdbc: An unsupervised multivariate analysis model for fraud detection in health insurance claims. *Expert Systems with Applications* **215**, 119259 (2023) <https://doi.org/10.1016/j.eswa.2022.119259>
- [9] Maiano, L., Montuschi, A., Caserio, M., Ferri, E., Kieffer, F., Germanò, C., Baiocco, L., Celsi, L.R., Amerini, I., Anagnostopoulos, A.: A deep-learning-based antifraud system for car-insurance claims. *Expert Systems with Applications* **231**, 120644 (2023) <https://doi.org/10.1016/j.eswa.2023.120644>
- [10] Zanke, P.: Ai-driven fraud detection systems: a comparative study across banking, insurance, and healthcare. *Advances in Deep Learning Techniques* **3**(2), 1–22 (2023)
- [11] Bello, O.A., Folorunso, A., Onwuchekwa, J., Ejiofor, O.E., Budale, F.Z., Egwuonwu, M.N., *et al.*: Analysing the impact of advanced analytics on fraud detection: A machine learning perspective. *European Journal of Computer Science and Information Technology* **11**(6), 103–126 (2023) <https://doi.org/10.37745/ejcsit.2013/vol11n6103126>
- [12] Mohammed, A.F.A., Rahman, H.M.A.-A.: The role of artificial intelligence (ai) on the fraud detection in the private sector in saudi arabia. *Journal of Arts, Literature, Humanities and Social Sciences* (100), 472–506 (2024) <https://doi.org/10.33193/JALHSS.100.2024.1018>
- [13] Yoo, Y., Shin, J., Kyeong, S.: Medicare fraud detection using graph analysis: a comparative study of machine learning and graph neural networks. *IEEE Access* (2023) <https://doi.org/10.1109/ACCESS.2023.3305962>
- [14] Zhang, C., Xiao, X., Wu, C.: Medical fraud and abuse detection system based on machine learning. *International Journal of Environmental Research and Public Health* **17**(19), 7265 (2020) <https://doi.org/10.3390/ijerph17197265>
- [15] Yang, C.C.: Explainable artificial intelligence for predictive modeling in healthcare. *Journal of healthcare informatics research* **6**(2), 228–239 (2022) <https://doi.org/10.1007/s41666-022-00114-1>
- [16] Angelov, P.P., Soares, E.A., Jiang, R., Arnold, N.I., Atkinson, P.M.: Explainable artificial intelligence: an analytical review. *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery* **11**(5), 1424 (2021)
- [17] Minh, D., Wang, H.X., Li, Y.F., Nguyen, T.N.: Explainable artificial intelligence: a comprehensive review. *Artificial Intelligence Review*, 1–66 (2022)
- [18] Došilović, F.K., Brčić, M., Hlupić, N.: Explainable artificial intelligence: A survey. In: 2018 41st International Convention on Information and Communication Technology, Electronics and Microelectronics (MIPRO), pp. 0210–0215 (2018). IEEE
- [19] Mill, E.R., Garn, W., Ryman-Tubb, N.F., Turner, C.: Opportunities in real time fraud detection: An explainable artificial intelligence (xai) research agenda. *International Journal of Advanced Computer Science and Applications* **14**(5), 1172–1186 (2023) <https://doi.org/10.1080/08839514.2024.2318670>
- [20] Abdul Kadhar, K.M., Anand, G., Abdul Kadhar, K.M., Anand, G.: Basics of python programming. *Data Science with Raspberry Pi: Real-Time Applications Using a Localized Cloud*, 13–47 (2021)
- [21] Brownlee, J.: Develop deep learning models on theano and tensorflow using keras. *J Chem Inf Model* **53**(9), 1689–1699 (2019)

- [22] Nguyen, Q.H., Ly, H.-B., Ho, L.S., Al-Ansari, N., Le, H.V., Tran, V.Q., Prakash, I., Pham, B.T.: Influence of data splitting on performance of machine learning models in prediction of shear strength of soil. *Mathematical Problems in Engineering* **2021**(1), 4832864 (2021)
- [23] Medicare Fraud Detection Dataset. <https://data.world/gymprathap/medicare-fraud-detection-dataset>
- [24] Famili, A., Shen, W.-M., Weber, R., Simoudis, E.: Data preprocessing and intelligent data analysis. *Intelligent Data Analysis* **1**(1), 3–23 (1997) [https://doi.org/10.1016/S1088-467X\(98\)00007-9](https://doi.org/10.1016/S1088-467X(98)00007-9)
- [25] Alasadi, S.A., Bhaya, W.S.: Review of data preprocessing techniques in data mining. *Journal of Engineering and Applied Sciences* **12**(16), 4102–4107 (2017) <https://doi.org/10.3923/jeasci.2017.4102.4107>
- [26] Kuzudisli, C., Bakir-Gungor, B., Bulut, N., Qaqish, B., Yousef, M.: Review of feature selection approaches based on grouping of features. *PeerJ* **11**, 15666 (2023) <https://doi.org/10.7717/peerj.15666>
- [27] Elreedy, D., Atiya, A.F.: A comprehensive analysis of synthetic minority oversampling technique (smote) for handling class imbalance. *Information Sciences* **505**, 32–64 (2019)
- [28] Agatonovic-Kustrin, S., Beresford, R.: Basic concepts of artificial neural network (ann) modeling and its application in pharmaceutical research. *Journal of pharmaceutical and biomedical analysis* **22**(5), 717–727 (2000)
- [29] Zupan, J.: Introduction to artificial neural network (ann) methods: what they are and how to use them. *Acta Chimica Slovenica* **41**(3), 327 (1994)
- [30] Dongare, A., Kharde, R., Kachare, A.D., *et al.*: Introduction to artificial neural network. *International Journal of Engineering and Innovative Technology (IJEIT)* **2**(1), 189–194 (2012)
- [31] Sharma, S., Sharma, S., Athaiya, A.: Activation functions in neural networks. *Towards Data Sci* **6**(12), 310–316 (2017) <https://doi.org/10.33564/IJEAST.2020.v04i12.054>
- [32] Ketkar, N., Moolayil, J., Ketkar, N., Moolayil, J.: Convolutional neural networks. *Deep learning with Python: learn best practices of deep learning models with PyTorch*, 197–242 (2021)
- [33] Graves, A., Graves, A.: Long short-term memory. *Supervised sequence labelling with recurrent neural networks*, 37–45 (2012)
- [34] Schmidhuber, J., Hochreiter, S., *et al.*: Long short-term memory. *Neural Comput* **9**(8), 1735–1780 (1997) <https://doi.org/10.1162/neco.1997.9.8.1735>
- [35] Buckland, M., Gey, F.: The relationship between recall and precision. *Journal of the American society for information science* **45**(1), 12–19 (1994) [https://doi.org/10.1002/\(SICI\)1097-4571](https://doi.org/10.1002/(SICI)1097-4571)
- [36] Yacoub, R., Axman, D.: Probabilistic extension of precision, recall, and f1 score for more thorough evaluation of classification models. In: *Proceedings of the First Workshop on Evaluation and Comparison of NLP Systems*, pp. 79–91 (2020)
- [37] Bettstetter, C., Hartenstein, H., Perez-Costa, X.: Stochastic properties of the random waypoint mobility model: epoch length, direction distribution, and cell change rate. In: *Proceedings of the 5th ACM International Workshop on Modeling Analysis and Simulation of Wireless and Mobile Systems*, pp. 7–14 (2002)
- [38] Wang, X., Jin, Y., Schmitt, S., Olhofer, M.: Recent advances in bayesian optimization. *ACM Computing Surveys* **55**(13s), 1–36 (2023)
- [39] Gelbart, M.A., Snoek, J., Adams, R.P.: Bayesian optimization with unknown constraints. *arXiv preprint arXiv:1403.5607* (2014)

- [40] Sheela, K.G., Deepa, S.N.: Review on methods to fix number of hidden neurons in neural networks. *Mathematical problems in engineering* **2013**(1), 425740 (2013)
- [41] Yuan, H., Xiong, F., Huai, X.: A method for estimating the number of hidden neurons in feed-forward neural networks based on information entropy. *Computers and Electronics in Agriculture* **40**(1-3), 57–64 (2003)
- [42] Aldughayfiq, B., Ashfaq, F., Jhanjhi, N., Humayun, M.: Explainable ai for retinoblastoma diagnosis: interpreting deep learning models with lime and shap. *Diagnostics* **13**(11), 1932 (2023)
- [43] Ng, C.H., Abuwala, H.S., Lim, C.H.: Towards more stable lime for explainable ai. In: 2022 International Symposium on Intelligent Signal Processing and Communication Systems (ISPACS), pp. 1–4 (2022). IEEE
- [44] Gaspar, D., Silva, P., Silva, C.: Explainable ai for intrusion detection systems: Lime and shap applicability on multi-layer perceptron. *IEEE Access* (2024)
- [45] Gupta, R.Y., Mudigonda, S.S., Baruah, P.K.: A comparative study of using various machine learning and deep learning-based fraud detection models for universal health coverage schemes. *International Journal of Engineering Trends and Technology* **69**(3), 96–102 (2021) <https://doi.org/10.14445/22315381/IJETT-V69I3P216>
- [46] Bounab, R., Zarour, K., Guelib, B., Khelifa, N.: Enhancing medicare fraud detection through machine learning: Addressing class imbalance with smote-enn. *IEEE Access* (2024) <https://doi.org/10.1109/ACCESS.2024.3385781>